Module 7: Molecular Genetics - Keys to Success

Learning Objectives

By the end of this module, you should be able to:

1. Describe DNA structure and base-pairing rules.
2. Explain why DNA replication is semiconservative.
3. Compare transcription and translation.
4. Use codons to connect mRNA sequence to amino acid sequence.
5. Explain how mutation can alter protein and trait outcomes.

Topics

- DNA structure
- Replication
- Transcription
- Translation
- Mutation and gene expression evidence

Key Terms to Know

- **DNA** - Nucleic acid that stores hereditary information.
- **Nucleotide** - DNA or RNA subunit with sugar, phosphate, and base.
- **Replication** - Process that copies DNA.
- **Transcription** - Synthesis of RNA from a DNA template.
- **Translation** - Synthesis of protein from mRNA codons.
- **Codon** - Three-base mRNA sequence read during translation.
- **Mutation** - Change in DNA sequence.
- **Gene expression** - Use of genetic information to make RNA or protein.

Core Contents

1. DNA stores information in nucleotide sequences and complementary base pairing.
2. Replication copies DNA using each strand as a template.
3. Transcription produces RNA from a DNA gene.
4. Translation uses mRNA codons and ribosomes to build proteins.
5. Mutations and expression patterns connect molecular changes to traits.

Study Tips

1. Start with the module's central claim before memorizing vocabulary.
2. Use the same-numbered lab as the evidence check for the module.
3. Explain one mechanism in words, then redraw it as a simple diagram.
4. Answer retrieval questions without notes, then revise with the terms list.

Connected Lab

- `lab-07_molecular-genetics.md`

Generated Visuals

- Module 07: Molecular Genetics Concept Map
- Module 07: Molecular Genetics Process Model
- Module 07: Molecular Genetics Retrieval Card